

Offline Flyback SMPS — Design Session Summary

1. Objective

Design, build, and validate an offline isolated flyback converter using a custom hand-wound transformer and a TinySwitch (TNY285) controller. Initial target was 5.0 V at ~0.3 A, with a future PCB target of 5.0 V at 1 A. No auxiliary winding was used.

2. Topology & Control

- Controller: TNY285PG
- Biasing: Internal drain-derived bias (no aux winding)
- Feedback: TL431 shunt regulator + optocoupler (LTV-817)
- Control pin: EN/UV pulled down by opto transistor
- Regulation method: Primary on-time modulation

3. Transformer

- Fully custom hand-wound HF transformer
- Correct polarity and turns on first attempt
- Converter regulated immediately on first full assembly
- Confirms magnetizing inductance and leakage were within acceptable limits

4. Primary Power Path & Protection

- 120 VAC → bridge → ~170 VDC bulk
- π filter: 47 μ F / 400 V – 1 mH – 47 μ F / 400 V
- Input protection: MOV + 0.5 A slow-blow fuse
- Clamp: RCD + Z (UF4007, 2.2 nF / 2 kV, 220 k Ω / 1 W, 47 Ω / 5 W, SMBJ250A)
- One IC failure occurred only when clamp was omitted

5. Secondary Power Path

- Rectifier: DFLS240-7 (40 V, 2 A Schottky)
- Output bulk: 470 μ F / 16 V (Panasonic FC)
- Output π filter: 1 mH inductor + additional bulk capacitance
- Optional ceramic bypass at rectifier node

6. Feedback Loop

- Divider used: 2.21 k Ω / 2.21 k Ω (equal-value)
- Resulting output: ~5.0 V
- Divider current: ~1.13 mA
- Opto LED resistor: 1 k Ω
- Compensation: 1 nF from TL431 cathode to REF
- Loop stable from no-load to full load

7. Secondary Snubber

- RC snubber across secondary diode
- Initial values: 3.3 nF (COG) + 3.3 Ω
- Intended to reduce diode recovery spikes and HF ringing

8. Measurements

- Output regulation: 5.00 V across all tested loads
- Drain measured with 500 $\times$ differential probe
- Bulk voltage: $\sim$ 170 VDC
- Peak VDS: $\sim$ 340–345 V
- Ringing frequency: $\sim$ 340–390 kHz
- Large margin to MOSFET rating

9. Output Noise Analysis

- HF spikes attributed to diode recovery + leakage resonance
- mV-scale ripple attributed to energy packets and limited output filtering
- Mitigations identified: snubber, increased bulk capacitance, π filter

10. Efficiency Measurement Plan

- Preferred method: true AC input power vs DC output power
- Secondary method: DC bus power estimation
- Goal: efficiency vs load curve

11. Documentation & BOM

- Full as-built BOM created with MPNs and Digi-Key PNs
- Output inductor and Z clamp added after review
- TL431 divider quick-reference sheet added
- BOM now PCB-ready and review-safe

12. Key Outcomes

- First full assembly regulated immediately
- Custom transformer successful on first attempt
- Clamp design validated by measurement
- Divider behavior fully understood
- Design process proven repeatable

13. Forward Work

- Middlebrook loop injection planned
- PCB Rev-A (5 V / 1 A)
- Next design: UCC-based 20 V, 5 A+ flyback